

Semi-simplicity of Tensor Products in Characteristic p

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Abstract. Let G be an affine group scheme over a field k of characteristic $p \neq 0$, and let (V_i) be a finite family of semi-simple representations of G . We show that if

$$\sum_i (\dim V_i - 1) < p,$$

then the tensor-product representation of G formed from the V_i is again semi-simple. Under the additional hypothesis that G is smooth, this theorem was proved by J.-P. Serre in 1994. We shall reduce to that case. More generally, the V_i may be taken to be objects of a tannakian category over k .

- 0. Introduction
- 1. Proof of Theorem 0.7
- 2. Saturation
- 3. The case where k is algebraically closed
- 4. Extension of scalars in a tannakian category, and proof of Theorem 0.1
- 5. Tensor products of exterior powers

0. Introduction

Let $\mathcal{T}$ be a tannakian category over a field k of characteristic $p \neq 0$. For the definition of tannakian categories, we refer to [3] p. 193 or to [1] 2.1, 2.8. Recall that the axioms include the existence of duals and of a fibre functor ω with values in vector spaces over a suitable extension K of k . The dimension $\dim(V)$ of an object V of $\mathcal{T}$ is $\dim_K \omega(V)$. Since two fibre functors become isomorphic over a sufficiently large suitable extension ([1] 1.12, 1.13), this dimension is independent of the choice of the fibre functor ω .

Theorem (0.1). *Let $(V_i)_{i \in I}$ be a finite family of objects of $\mathcal{T}$. If the V_i are semi-simple and*

$$\sum_{i \in I} (\dim(V_i) - 1) < p,$$

then the tensor product of the V_i is a semi-simple object of $\mathcal{T}$.

Example 0.2. If $\mathcal{T}$ is the category of finite-dimensional linear representations of an affine group scheme over k , one obtains the theorem announced in the abstract.

Other examples are the category of finite-dimensional representations of a Lie algebra over k , or that of finite-dimensional p -representations of a p -Lie algebra over k . By passage to an algebraic envelope, these cases in fact reduce to the preceding one.

Example 0.3. Suppose that a group Γ acts on a commutative field K of characteristic p , and let $k := K^\Gamma$ be the subfield of invariants. Let $\mathcal{T}$ be the category of finite-dimensional K -vector spaces endowed with a semi-linear action of Γ . With the tensor product over K , this category is tannakian over k : it admits the fibre functor over K “underlying vector space”. If semi-linear representations V_i of Γ on K are semi-simple and

$$\sum_i (\dim_K(V_i) - 1) < p,$$

then the semi-linear tensor-product representation of the V_i is semi-simple.

0.4. Let V be a finite-dimensional vector space over k , of characteristic p . As explained in [4] §4, an automorphism g of V such that $g^p = 1$ defines a morphism of algebraic groups

$$\mathbb{G}_a \longrightarrow \mathrm{GL}(V) : \quad t \longmapsto g^t,$$

where g^t is defined by the binomial formula

$$g^t := \sum_{n < p} \binom{t}{n} (g - 1)^n. \tag{0.4.1}$$

Similarly, an endomorphism N of V such that $N^p = 0$ defines a morphism of algebraic groups

$$\mathbb{G}_a \longrightarrow \mathrm{GL}(V) : \quad t \longmapsto \exp(tN),$$

where $\exp(tN)$ is defined by

$$\exp(tN) := \sum_{n < p} \frac{(tN)^n}{n!}. \quad (0.4.2)$$

Indeed, after any scalar extension to a commutative k -algebra Λ , if for $t \in \Lambda$ one defines $\exp(tN)$ by (0.4.2), the hypothesis $N^p = 0$ ensures that

$$\exp((t' + t'')N) = \exp(t'N) \exp(t''N).$$

Definition 0.5. Suppose that k is algebraically closed of characteristic p , and let G be a subgroup scheme of $\mathrm{GL}(V)$.

- (i) G is *saturated* if, for every $g \in G(k)$ such that $g^p = 1$, the morphism $t \mapsto g^t$ from $\mathbb{G}_a$ to $\mathrm{GL}(V)$ factors through G .
- (ii) G is *infinitesimally saturated* if, for every $N \in \mathrm{Lie}(G)$ such that $N^p = 0$ (in the sense of the p -Lie-algebra structure on $\mathrm{Lie}(G)$, or as an endomorphism of V ; this is the same condition), the morphism $t \mapsto \exp(tN)$ from $\mathbb{G}_a$ to $\mathrm{GL}(V)$ factors through G .

We shall say that G is *doubly saturated* if it is saturated and infinitesimally saturated.

0.6. Let G be an affine group scheme of finite type over an algebraically closed field k of characteristic p . In the statement of 0.1, take $\mathcal{T}$ to be the category of finite-dimensional representations of G , assume that the family (V_i) of representations is faithful, and put $V := \bigoplus V_i$. To prove 0.1 when G is reduced, Serre [4] first reduces to the case where G is saturated in $\mathrm{GL}(V)$. If this is the case, the finite group $\pi_0(G)$ of connected components of G has order prime to p , and this permits one to reduce to the case where G is connected. For G not necessarily reduced, we use an analogous argument to reduce to the case where G is infinitesimally saturated in $\mathrm{GL}(V)$, and our key result in that case is the following structure theorem 0.7. It will give a reduction to the case where G is smooth.

Let G^0 be the neutral component of G , G_{red}^0 the reduced subgroup scheme of G^0 , and $R_u G_{\mathrm{red}}^0$ the unipotent radical of G_{red}^0 . Both G_{red}^0 and $R_u G_{\mathrm{red}}^0$ are smooth, i.e. linear algebraic groups in the sense of the Chevalley seminar of 1956/58.

Theorem (0.7). *Let $(V_i)_{i \in I}$ be a finite faithful family of representations of G . Let V be the sum of the V_i . Assume that each representation V_i has dimension $< p$ and that, as a subgroup of $\mathrm{GL}(V)$, G is infinitesimally saturated. Under these hypotheses:*

- (i) G_{red}^0 and $R_u G_{\mathrm{red}}^0$ are invariant subgroup schemes of G , and G^0/G_{red}^0 is of multiplicative type.
- (ii) If the representations V_i are semi-simple, G_{red}^0 is reductive.
- (iii) If G_{red}^0 is reductive, there exists in G^0 a connected central subgroup scheme M of multiplicative type such that the morphism

$$M \times G_{\mathrm{red}}^0 \longrightarrow G^0$$

makes G^0 a quotient of $M \times G_{\mathrm{red}}^0$.

1. Proof of Theorem 0.7

Proposition (1.1). *In an affine group scheme G of finite type over a field, every connected subgroup scheme of multiplicative type is contained in a maximal connected subgroup scheme of multiplicative type.*

Proof. (after a letter from M. Raynaud to the author). It is enough to show that, in the set of connected subgroup schemes of multiplicative type of G , ordered by inclusion, every totally ordered subset $\mathcal{M}$ has an upper bound (Bourbaki, Ens III §2 no4 Theorem 2). For $M \in \mathcal{M}$, the centralizer $Z_G(M)$ of M in G is a subgroup scheme (automatically closed), and the map $M \mapsto Z_G(M)$ is decreasing. The ordered set of closed subgroup schemes of G is noetherian. The set of the $Z_G(M)$, for $M \in \mathcal{M}$, therefore has a smallest element C . Each $M \in \mathcal{M}$ is contained in the centre $Z(C)$ of C . This centre is affine and commutative. The neutral component of its largest subgroup scheme of multiplicative type (SGA3 XVII 7.2.1) majorizes $\mathcal{M}$. This is the required upper bound. $\square$

1.2. Fix k , G , and the finite family of representations $(V_i)_{i \in I}$, with sum V , satisfying the hypotheses of 0.7: k is algebraically closed of characteristic $p \neq 0$, $\dim(V_i) < p$, and, as a subgroup scheme of $\mathrm{GL}(V)$, G is infinitesimally saturated.

Lemma (1.3). (i) *The neutral component G^0 of G is infinitesimally saturated in $\mathrm{GL}(V)$.*

(ii) *If a representation W of G is semi-simple, its restriction to G^0 is also semi-simple.*

(iii) *If the reduced subgroup G_{red}^0 of G^0 (respectively its unipotent radical $R_u G_{\mathrm{red}}^0$) is an invariant subgroup scheme of G^0 , then it is one of G .*

Proof. (i) follows from $\mathrm{Lie} G = \mathrm{Lie} G^0$ and from the connectedness of $\mathbb{G}_a$.

It is enough to prove (ii) for W irreducible. Let $S \subset W$ be the sum of the irreducible G^0 -subrepresentations of W . We have to show that $S = W$. Each $g \in G(k)$ normalizes G^0 . By transport of structure, S is therefore stable under g . Since k is algebraically closed, G is the union of the gG^0 for $g \in G(k)$, and S is stable under G . Since W is irreducible and $S \neq 0$, S is all of W .

Similarly, by transport of structure, each $g \in G(k)$ normalizes G_{red}^0 and $R_u G_{\mathrm{red}}^0$, and (iii) follows from the fact that G is the union of the gG^0 . $\square$

1.4. Lemma 1.3 shows that, in order to prove 0.7 for G , it is enough to prove it for G^0 . Moreover, if $p = 2$, each V_i has dimension 1, G is of multiplicative type, and 0.7 is trivial.

In the rest of this section, in addition to the hypotheses of 1.2, we shall assume that $p \neq 2$ and that G is connected: $G = G^0$. We shall prove that G_{red} is an invariant subgroup scheme of G in 1.18, that G/G_{red} is of multiplicative type in 1.19, the invariance in G of $R_u G_{\mathrm{red}}$ in 1.20–1.23, 0.7 (ii) in 1.24, and 0.7 (iii) in 1.25.

1.5. Let H be a maximal connected subgroup of G of multiplicative type (1.1), and let $Z_G^0(H)$ be the neutral component of its centralizer $Z_G(H)$. The quotient

$$U := Z_G^0(H)/H$$

contains no subgroup μ_p , because the inverse image in $Z_G^0(H)$ of such a subgroup would be an extension of μ_p by H , hence of multiplicative type (SGA3 XVII 7.1.1 b)). The group U is therefore unipotent (SGA3 XVII 4.3.1 (v)).

1.6. Let X be the character group of H . The connectedness of H is equivalent to X having no prime-to- p torsion. Giving an action of H on a vector space W is equivalent to giving an X -grading

$$W = \bigoplus_{\alpha \in X} W^\alpha$$

of W , and this construction is functorial and compatible with tensor products. In particular, V and the V_i are X -graded, and the action of H on G by inner automorphisms defines an X -grading

$$\mathrm{Lie}(G) = \bigoplus_{\alpha \in X} \mathrm{Lie}(G)^\alpha$$

of $\mathrm{Lie}(G)$.

Lemma (1.7). *The central extension $Z_G^0(H)$ of U by H is split.*

Since U is unipotent and H is of multiplicative type, there is no non-trivial morphism from U to H (SGA3 XVII 2.4 (ii)). The splitting guaranteed by Lemma 1.7 is therefore unique. We shall use the corresponding isomorphism of $H \times U$ with $Z_G^0(H)$ to regard U as a subgroup of $Z_G^0(H)$.

Proof. Let V_i^α ($\alpha \in X$) be the homogeneous components of the V_i . The group $Z_G^0(H)$ acts on V_i^α . Let χ_i^α be the determinant character of $Z_G^0(H)$ attached to this representation. Its restriction to H is $\dim(V_i^\alpha)$ times the character α of H .

Let $J \subset I \times X$ be the set of (i, α) such that $V_i^\alpha \neq 0$, and for $j = (i, \alpha)$, put $V^j := V_i^\alpha$, $\chi_j := \chi_i^\alpha$. The family (V_i) of representations of H is faithful: the α 's for $j = (i, \alpha) \in J$ generate X . If $j = (i, \alpha)$ is in J , then $\dim V_i^\alpha \leq \dim V_i < p$, hence it is prime to p . The restrictions to H of the χ_j ($j \in J$) therefore generate a subgroup of X of finite index prime to p . Let F be the kernel of

$$(\chi_j) : H \longrightarrow \mathbb{G}_m^J$$

and let Q be the cokernel. The group F is finite, reduced, and of order prime to p . In the commutative diagram with exact rows

$$\begin{array}{ccccccc} 1 & \longrightarrow & H/F & \longrightarrow & Z_G^0(H)/F & \longrightarrow & U \longrightarrow 1 \\ & & \parallel & & \downarrow (\chi_j) & & \downarrow \\ 1 & \longrightarrow & H/F & \xrightarrow{(\chi_j)} & \mathbb{G}_m^J & \longrightarrow & Q \longrightarrow 1, \end{array}$$

the morphism from the unipotent group U to the group of multiplicative type Q is trivial. The kernel of (χ_j) therefore lifts U into $Z_G^0(H)/F$. Its inverse image in $Z_G^0(H)$ is a central extension E of U by F . By Lemma 1.8 below, this extension is trivial. A lifting of U into E splits the central extension $Z_G^0(H)$ of U by H . $\square$

Lemma (1.8). *A central extension of a unipotent group by a finite (reduced) group of order prime to p is trivial.*

We shall give two proofs of this lemma, based on different ideas. The second assumes U connected, a case sufficient for the application above.

First proof. Let $1 \rightarrow F \rightarrow E \rightarrow U \rightarrow 1$ be a central extension as in the lemma, and let a be an integer prime to p which kills F . Regard E as a central extension in the category of fppf sheaves of groups on $\text{Spec}(k)$. For any S/k , $U(S)$ admits a finite central filtration with successive quotients killed by p (apply SGA3 XV 3.5 (v)). Every finite subset X of $U(S)$ therefore generates a finite p -group P_X . Let Q_X be its intersection with the image of $E(S)$. The inverse image of Q_X in $E(S)$ is a central extension E_X of Q_X by $F(S) = F^{\pi_0(S)}$. The group $H^2(Q_X, F(S))$, which classifies these extensions, is trivial, since it is killed both by a and by the power $|Q_X|$ of p . The extension E_X is therefore trivial. Moreover, the lifting of Q_X into E_X is unique: it is the set of a -th powers of elements of E_X .

Passing to the filtered direct limit over X and then sheafifying, one obtains that the subsheaf of sets of E which is the image of $x \mapsto x^a : E \rightarrow E$ is a subsheaf of groups lifting U in E . $\square$

Second proof. The action of F on E by right translations makes E an F -torsor over U . It is trivialized above the neutral element, and hence corresponds to a morphism

$$\pi_1(U, e) \longrightarrow F.$$

The scheme U_{red} is isomorphic to an affine space $\mathbb{A}^N$. Since

$$\pi_1(U, e) = \pi_1(U_{\text{red}}, e) \simeq \pi_1(\mathbb{A}^N, 0)$$

has no non-trivial quotient of order prime to p (SGA4 XV 2.2), the F -torsor E is trivial: the neutral component of E lifts U in E . $\square$

Lemma (1.9). *Let $f : V \rightarrow W$ be a morphism of schemes of finite type over k , let $v \in V(k)$, and let $(df)_v$ be the morphism induced by f from the Zariski tangent space of V at v to the Zariski tangent space of W at $f(v)$. If V is smooth over k and $(df)_v$ is an isomorphism, then f is étale at v , and W is smooth over k in a neighbourhood of $f(v)$.*

Proof. Let A (respectively B) be the completion of the local ring of V at v (respectively of W at $f(v)$). Let $\mathfrak{m}$ be the maximal ideal of A , and $\mathfrak{n}$ that of B . Injectivity of $(df)_v$ ensures that A is a quotient of B . We have to show that $B \xrightarrow{\sim} A$ (SGA1 I 4.4). Let $t_1, \dots, t_N \in \mathfrak{n}$ lift a basis of $\mathfrak{n}/\mathfrak{n}^2$. The bijectivity of $(df)_v$ ensures that the images of the t_i in $\mathfrak{m}/\mathfrak{m}^2$ form a basis. The t_i define an epimorphism

$$g : T_i \longmapsto t_i : k[[T_1, \dots, T_N]] \longrightarrow B.$$

In the commutative diagram

$$\begin{array}{ccc} & k[[T_1, \dots, T_N]] & \\ \swarrow & & \searrow g \\ B & \xrightarrow{\quad} & A, \end{array}$$

the smoothness of V at v ensures that g is an isomorphism. It follows that $B \xrightarrow{\sim} A$. $\square$

In characteristic p , the Lie algebra of a group scheme is a p -Lie algebra. We write $\gamma \mapsto \gamma^p$ for the operation of taking the p -th power, often denoted $\gamma^{[p]}$. For every linear representation ρ of the group, one has $\rho(\gamma^p) = \rho(\gamma)^p$.

Lemma (1.10). *For every $\gamma \in \text{Lie } U$, one has $\gamma^p = 0$.*

As explained after 1.7, U is here regarded as a subgroup of $Z_G^0(H) \subset G$.

Proof. Since U is a unipotent group scheme, for each i the image γ_i of γ in $\text{End}(V_i)$ is nilpotent. Since $\dim(V_i) < p$, one has $\gamma_i^p = 0$. As the family (V_i) of representations is faithful, $\gamma^p = 0$. $\square$

Corollary (1.11). *The unipotent group U is smooth.*

Proof. Let $(\gamma_a)_{1 \leq a \leq A}$ be a basis of $\text{Lie } U$. By 1.10, for each a ,

$$\rho_a : \lambda \mapsto \exp(\lambda \gamma_a) : \mathbb{G}_a \longrightarrow \text{GL}(V)$$

is defined (0.4). The hypothesis that G is infinitesimally saturated ensures that this morphism factors through G . Since U centralizes H , the action of H by inner automorphisms fixes $\text{Lie } U$, hence each ρ_a , and ρ_a factors through $Z_G(H)$, indeed through $Z_G^0(H) = H \times U$ because $\mathbb{G}_a$ is connected. Since every morphism from $\mathbb{G}_a$ to H is trivial, ρ_a factors through U . Let $\mathbb{E}^A$ be affine A -space and

$$f : \mathbb{E}^A \longrightarrow U : (t_a) \mapsto \prod \rho_a(t_a),$$

where the product is taken with respect to the group law of U . By construction, df is an isomorphism at 0. Applying 1.9, one obtains that U is smooth in a neighbourhood of the neutral element, hence everywhere by homogeneity. $\square$

Recall that the action of H on $\text{Lie } G$ by inner automorphisms defines a grading

$$\text{Lie}(G) = \bigoplus_{\alpha \in X} \text{Lie}(G)^\alpha.$$

Lemma (1.12). *If $\alpha \in X$ is non-zero and $\gamma \in \text{Lie}(G)^\alpha$, then $\gamma^p = 0$.*

Proof. Regard γ as an endomorphism of V . If γ^p were non-zero—whether in the sense of the p -Lie-algebra structure on $\text{Lie } G$ or as an endomorphism of V , which is the same here—there would exist $i, \beta \in X$, and $v \in V_i^\beta$ such that $\gamma^p v \neq 0$. This implies the non-vanishing of the $\gamma^l v$ for $0 \leq l \leq p$. The element $\gamma^l v$ of V_i lies in $V_i^{\beta+l\alpha}$. For $0 \leq l < p$, the characters $\beta + l\alpha$ are distinct, because X has no prime-to- p torsion. The vectors $\gamma^l v$, $0 \leq l < p$, are therefore linearly independent. This contradicts the hypothesis $\dim V_i < p$. $\square$

1.13. Let $(\gamma_b)_{1 \leq b \leq B}$ be a family of elements of $\text{Lie } G$, a disjoint union of bases of the $(\text{Lie } G)^\alpha$ for $\alpha \neq 0$. Let α_b denote the character α of H such that $\gamma_b \in (\text{Lie } G)^\alpha$. Since G is infinitesimally saturated, the morphism 0.4

$$\lambda \mapsto \exp(\lambda \gamma_b) : \mathbb{G}_a \longrightarrow \text{GL}(V)$$

given by 1.12 factors through a morphism

$$\rho_b : \mathbb{G}_a \longrightarrow G.$$

This morphism is equivariant for the action of H on $\mathbb{G}_a$ by α_b and on G by inner automorphisms, because the same equivariance holds for its composite with the inclusion of G in $\text{GL}(V)$.

Let q (respectively q_0) be the projection of G onto $G/Z_G(H)$ (respectively $G/Z_G^0(H)$). Consider the morphism

$$f : \mathbb{E}^B \longrightarrow G : (t_b) \mapsto \prod_1^B \rho_b(t_b),$$

where the product is taken with respect to the group law of G .

Lemma (1.14). *The composite morphism $qf : \mathbb{E}^B \rightarrow G/Z_G(H)$ is étale at 0.*

Since $G/Z_G^0(H)$ is étale over $G/Z_G(H)$, the same statement holds for $q_0 f$.

Proof. Let e be the neutral element of G , and let $T_{q(e)}$ be the tangent space of $G/Z_G(H)$ at $q(e)$. By 1.9, it is enough to verify that qf induces an isomorphism from the tangent space at 0 to $T_{q(e)}$, and for this it is enough to verify that $dq : \text{Lie}(G) \rightarrow T_{q(e)}$ induces an isomorphism

$$\bigoplus_{\alpha \neq 0} \text{Lie}(G)^\alpha \xrightarrow{\sim} T_{q(e)}.$$

If G were smooth, $Z_G(H)$ would be smooth, as the fixed locus of an action on G of the group H of multiplicative type; one would obtain an exact sequence

$$0 \longrightarrow \mathrm{Lie} Z_G^0(H) \longrightarrow \mathrm{Lie} G \longrightarrow T_{q(e)} \longrightarrow 0$$

and would conclude by observing that $\mathrm{Lie} Z_G^0(H) = \mathrm{Lie}(G)^H$.

We shall deduce 1.14 from the following lemma. □

Lemma (1.15). *Suppose that a group H of multiplicative type, with character group X , acts on an affine k -scheme of finite type $S = \mathrm{Spec}(A)$. Let S^H be the fixed locus and $s \in S^H(k)$. Let I be the ideal defining S^H in S . Regard I/I^2 as a coherent sheaf on S^H , and let $(I/I^2)_s = s^*(I/I^2)$ be its fibre at s . Let $\mathfrak{m}$ be the maximal ideal at s . The group H acts on $\mathfrak{m}/\mathfrak{m}^2$, which is therefore X -graded. The inclusion of I in $\mathfrak{m}$ induces an isomorphism*

$$(I/I^2)_s \xrightarrow{\sim} \bigoplus_{\alpha \neq 0} (\mathfrak{m}/\mathfrak{m}^2)^\alpha. \quad (1.15.1)$$

The hypothesis that S is affine is probably unnecessary.

Proof. The action of H gives an X -grading of A . The quotient A/I is the largest graded quotient of A concentrated purely in degree 0: I is the ideal generated by the A^α for $\alpha \neq 0$. Both sides of (1.15.1) are unchanged when A is replaced by A/I^2 . We may therefore suppose that $I^2 = 0$. Then $A^\alpha A^\beta = 0$ for $\alpha, \beta \neq 0$, I is the sum of the A^α for $\alpha \neq 0$, and

$$A^0 \xrightarrow{\sim} A/I.$$

Let $\mathfrak{m}_0$ be the maximal ideal of s in $A/I \xleftarrow{\sim} A^0$. We have

$$\mathfrak{m} = \mathfrak{m}_0 \oplus \bigoplus_{\alpha \neq 0} A^\alpha.$$

Hence

$$\mathfrak{m}/\mathfrak{m}^2 = \mathfrak{m}_0/\mathfrak{m}_0^2 \oplus \bigoplus_{\alpha \neq 0} A^\alpha \otimes_{A^0} A^0/\mathfrak{m}_0,$$

$$I/I^2 \otimes_{A^0} A^0/\mathfrak{m}_0 = \bigoplus_{\alpha \neq 0} A^\alpha \otimes_{A^0} A^0/\mathfrak{m}_0,$$

and (1.15.1) follows. □

End of the proof of 1.14. The morphism $G \rightarrow G/Z_G(H)$ is flat. If I is the ideal defining $Z_G(H)$ in G , and $\mathfrak{n}$ is the maximal ideal at the point $q(e)$ of $G/Z_G(H)$, the bundle I/I^2 on $Z_G(H)$ is the inverse image of $\mathfrak{n}/\mathfrak{n}^2$ on $q(e)$. Its fibre at the neutral element e of G is $\mathfrak{n}/\mathfrak{n}^2$, and by (1.15.1) we therefore have

$$\mathfrak{n}/\mathfrak{n}^2 \xrightarrow{\sim} \bigoplus_{\alpha \neq 0} (\mathfrak{m}/\mathfrak{m}^2)^\alpha.$$

Passing to duals gives

$$\bigoplus_{\alpha \neq 0} \mathrm{Lie}(G)^\alpha \xrightarrow{\sim} T_{q(e)}.$$

Corollary (1.16). *The morphism*

$$F : \mathbb{E}^B \times Z_G^0(H) \longrightarrow G : (t_b), z \longmapsto \prod \rho_b(t_b) \cdot z$$

is étale at $(0, e)$.

Proof. The square

$$\begin{array}{ccc} \mathbb{E}^B \times Z_G^0(H) & \xrightarrow{F} & G \\ \downarrow & & \downarrow \\ \mathbb{E}^B & \xrightarrow{q_0 f} & G/Z_G^0(H) \end{array}$$

is cartesian. The morphism F is therefore étale at every point lying above $0 \in \mathbb{E}^B$. $\square$

1.17. If $f : T \rightarrow U$ is a quasi-compact morphism of schemes, the closed schematic image $f(T)^-$ of f is the smallest closed subscheme of U through which f factors. If $f : \text{Spec}(A) \rightarrow \text{Spec}(B)$ is affine, it is defined by the kernel ideal of $f^* : B \rightarrow A$. Its formation is compatible with every flat base change $U' \rightarrow U$. If f is a morphism of schemes over S , if a group scheme H , flat over S , acts on T and U , and if f is equivariant, this compatibility with base change implies that the action of H on U induces an action of H on the closed schematic image $f(T)^-$.

1.18. Proof of the invariance of G_{red} in G . The morphism F of 1.16 is equivariant for the action of H on $\mathbb{E}^B$ by the α_b , and on G by inner automorphisms. Since U is smooth (1.11), the reduced subscheme of $Z_G^0(H) = H \times U$ is $H_{\text{red}} \times U$. An étale morphism induces an étale morphism on reduced subschemes. The morphism induced by F from $\mathbb{E}^B \times H_{\text{red}} \times U$ to G_{red} is therefore étale at 0, and its image contains a non-empty open subset of G_{red} . The closed schematic image of the morphism

$$F_1 : \mathbb{E}^B \times H_{\text{red}} \times U \longrightarrow G$$

induced by F therefore coincides with G_{red} . Since this morphism is H -equivariant, 1.17 ensures that G_{red} is stable under the action of H on G by inner automorphisms.

Because F is étale at the origin, G is generated as an fppf sheaf by the image of F , hence by H and G_{red} , since $U \subset G_{\text{red}}$. Since H normalizes G_{red} , it follows that G_{red} is an invariant subgroup of G .

Proposition (1.19). *One has*

$$H/H_{\text{red}} \xrightarrow{\sim} G/G_{\text{red}}.$$

Consequently, G/G_{red} is of multiplicative type.

Proof. If Y_1 and Y_2 are two closed subschemes of Z , then, for every morphism $u : Z' \rightarrow Z$,

$$u^{-1}(Y_1 \cap Y_2) = u^{-1}(Y_1) \cap u^{-1}(Y_2).$$

If u is étale and $Y_2 = Z_{\text{red}}$, this gives

$$u^{-1}(Y_1 \cap Z_{\text{red}}) = u^{-1}(Y_1) \cap Z'_{\text{red}}.$$

Apply this to F , which is étale in a neighbourhood of $\{0\} \times H \subset \{0\} \times Z_G^0(H)$. In a neighbourhood of $\{0\} \times H$, the inverse image of $H \cap G_{\text{red}}$ is therefore H_{red} , so that

$$H_{\text{red}} \longrightarrow H \cap G_{\text{red}}$$

is étale, hence an isomorphism. If $H, H_{\text{red}}, G, G_{\text{red}}$, and the quotients are interpreted as fppf-sheaves, the isomorphism

$$H/H_{\text{red}} \xrightarrow{\sim} G/G_{\text{red}}$$

follows from the facts that H normalizes G_{red} , that $HG_{\text{red}} = G$, and that $H_{\text{red}} \xrightarrow{\sim} H \cap G_{\text{red}}$. $\square$

1.20. Proof of the invariance in G of $R_u G_{\text{red}}$. Let R denote the unipotent radical $R_u G_{\text{red}}$ of the smooth group G_{red} , and let L be the quotient G_{red}/R . Let T_L be a maximal torus of the reductive group L . It is a maximal connected subgroup of multiplicative type. The inverse image of T_L in G_{red} is an extension of the torus T_L by the smooth connected unipotent group R . By SGA3 XVII 5.1.1 (i), such an extension is trivial: the torus T_L lifts to a torus T , automatically maximal, of G_{red} . We take H to be a maximal connected subgroup of multiplicative type in G containing T . Then

$$T = H_{\text{red}} = H \cap G_{\text{red}}.$$

Lemma (1.21). *The connected centralizer $Z_{G_{\text{red}}}^0(T)$ of T in G_{red} is a product $T \times W$, with W unipotent. The group scheme H normalizes W .*

Proof. Since H normalizes T , it normalizes $Z := Z_{G_{\text{red}}}^0(T)$. Like G , the subgroup G_{red} of $\text{GL}(V)$ is infinitesimally saturated. Applying 1.7 to G_{red} and T (instead of to G and H) gives the decomposition $Z = T \times W$. The proof of 1.7 describes $W \subset Z$ as the neutral component of the intersection of the kernels of the determinant characters of the representations of Z in the V_i^β , for β a weight of T in V . This description shows that H normalizes W . $\square$

Lemma (1.22). *The action of H on $\mathrm{Lie}(G_{\mathrm{red}})$ preserves the subspace $\mathrm{Lie}(R)$.*

Proof. Decompose $l = \mathrm{Lie} L$, $g = \mathrm{Lie} G_{\mathrm{red}}$, and $r = \mathrm{Lie} R$ according to the weights of T . Since H normalizes T , the resulting decomposition of g is respected by H . We shall show that each r_α , and hence their sum r , is stable under H .

The connected centralizer of T_L in L is just T_L . Thus one has a morphism of exact sequences

$$\begin{array}{ccccccc} 1 & \longrightarrow & W & \hookrightarrow & Z = T \times W & \longrightarrow & T_L \\ & & \downarrow & & \downarrow & & \downarrow \\ 1 & \longrightarrow & R & \hookrightarrow & G_{\mathrm{red}} & \longrightarrow & L. \end{array}$$

The weight-zero part of g is $\mathrm{Lie}(Z)$. Its intersection with $\mathrm{Lie} R$ is $\mathrm{Lie} W$, which is stable under H by 1.21.

If α is not a weight of T on l , then $r_\alpha = g_\alpha$, stable under H . If α is a non-zero weight of T on l , then l_α and $l_{-\alpha}$ are one-dimensional. Under our hypothesis $p \neq 2$, the bracket induces a non-degenerate pairing of l_α and $l_{-\alpha}$ with values in a line $h_\alpha \subset \mathrm{Lie} T_L$. The bracket of g induces a pairing of g_α and $g_{-\alpha}$ with values in $g_0 = \mathrm{Lie} Z$. Its composite with

$$g_0 \longrightarrow g_0 / \mathrm{Lie} W \simeq \mathrm{Lie} T_L$$

factors through $g_{\pm\alpha} \rightarrow l_{\pm\alpha}$ and the preceding pairing. It therefore takes values in a line d of $g_0 / \mathrm{Lie} W$, and

$$r_{\pm\alpha} = \mathrm{Ker}(g_{\pm\alpha} \longrightarrow \mathrm{Hom}(g_{\mp\alpha}, d)) :$$

the $r_{\pm\alpha}$ are the kernels of the pairing of g_α and $g_{-\alpha}$ with values in d . Since the decomposition of g according to the weights of T , and also $W \subset g_0$, are respected by H , this description shows that the r_α , and therefore r , are subrepresentations of H acting on g . $\square$

1.23. End of the proof of the invariance of R . Let Fil be the finite increasing filtration of $\bigoplus V_i$ such that $\mathrm{Fil}_0 = 0$ and $\mathrm{Fil}_{n+1} / \mathrm{Fil}_n$ is the space of R -invariants in $\bigoplus V_i / \mathrm{Fil}_n$. It is the sum of filtrations of the V_i . It is respected by G_{red} , since R is an invariant subgroup of G_{red} .

The reduced subgroup of the largest subgroup of G_{red} acting trivially on $\mathrm{Gr} \mathrm{Fil}(\bigoplus V_i)$ is a smooth invariant unipotent subgroup containing R . It therefore coincides with R .

If $\gamma \in \mathrm{Lie} R$, then γ acts nilpotently on V_i . Since the family of representations V_i , of dimension $< p$, is faithful, one has $\gamma^p = 0$. Since G_{red} is infinitesimally saturated, the corresponding morphism

$$\rho : \mathbb{G}_a \longrightarrow \prod_i \mathrm{GL}(V_i)$$

factors through G_{red} . As this representation of $\mathbb{G}_a$ is trivial on $\mathrm{Gr} \mathrm{Fil}(\bigoplus V_i)$, the morphism ρ even factors through R .

Choose a basis $(\gamma_c)_{1 \leq c \leq C}$ of $r = \mathrm{Lie} R$, the disjoint union of bases of the r_β , where β ranges through the weights of H acting on r . Let $\rho_c : \mathbb{G}_a \rightarrow R$ be the morphism defined by γ_c . As in 5.6, the morphism of schemes

$$f : \mathbb{E}^C \longrightarrow R, \quad (t_c) \longmapsto \prod_{c=1}^C \rho_c(t_c)$$

is étale at 0. Its composite with the inclusion of R into G_{red} is H -equivariant, for a suitable action of H on $\mathbb{E}^C$. By the same argument as in 1.17 and 1.18, it follows that R is stable under the action of H on G_{red} .

Corollary (1.24). *If the representations V_i are semisimple, the group scheme G_{red} is reductive.*

Proof. Replacing each V_i by a family of irreducible representations whose direct sum is V_i , one is reduced to assuming the V_i irreducible. We must prove that the unipotent radical $R = R_u G_{\mathrm{red}}$ is trivial.

Since R is unipotent, the subspace V_i^R of R -invariants in V_i is non-zero. Because R is invariant in G , V_i^R is stable under G ; hence, by irreducibility of V_i , it equals V_i . Since the family of V_i is faithful, R is trivial. $\square$

1.25. Proof of 0.7 (iii). Assume G_{red} is reductive, let T be a maximal torus of G_{red} , and take H to be a maximal connected subgroup of multiplicative type in G containing T . It normalizes G_{red} , and its action on G_{red}

fixes T . The group scheme of automorphisms of the reductive group G_{red} that fix T is the image T^{ad} of T in the adjoint group. If Z is the subgroup of H centralizing G_{red} , the morphism

$$Z \times T \longrightarrow H$$

is a quotient of T : the image T^{ad} of T in the adjoint group. Regard all these group schemes as fppf-sheaves. Since H and G_{red} generate G , Z is central. Since the action of H on G_{red} factors through a quotient of T , Z and T generate H . Since Z is generated by Z^0 and Z_{red} , and $Z_{\text{red}} \subset G_{\text{red}}$, Z^0 and G_{red} generate G : the morphism

$$Z^0 \times G_{\text{red}} \longrightarrow G$$

is an epimorphism, and this verifies 0.7 (iii).

2. Saturation (cf. [4] §4)

2.1. Let V be a finite-dimensional vector space over the algebraically closed field k of characteristic p , and let G be a subgroup scheme of $\text{GL}(V)$. The intersection G^* of the doubly saturated subgroup schemes (0.5) of $\text{GL}(V)$ containing G is doubly saturated. We shall call G^* the double saturation of G .

Let $(V_i)_{i \in I}$ be a finite family of finite-dimensional vector spaces over k . For each i , let N_i be an endomorphism of V_i with $N_i^p = 0$. It defines an action $t \mapsto \exp(tN_i)$ of $\mathbb{G}_a$ on V_i (0.4). For $a \in I$, write again N_a for the endomorphism of $\bigotimes_i V_i$ which is the tensor product of $N_a \in \text{End}(V_a)$ and the identities $1 \in \text{End}(V_j)$ for $j \neq a$. Write N for the endomorphism $\sum_a N_a$ of $\bigotimes_i V_i$.

Proposition (2.2). *Assume that there exist integers v_i such that $N_i^{v_i} = 0$ and $\sum_i (v_i - 1) < p$. Then $N^p = 0$, and the action $t \mapsto \exp(tN)$ of $\mathbb{G}_a$ on $\bigotimes_i V_i$ is the tensor product of the actions of $\mathbb{G}_a$ on the V_i defined by the N_i .*

Proof. The endomorphisms N_a of $\bigotimes_i V_i$ commute with one another, and every monomial of degree p in the N_a has degree $\geq v_a$ in N_a for at least one $a \in I$; it is therefore zero. This gives

$$N^p = \left(\sum_a N_a \right)^p = 0,$$

as well as the identity between truncated exponentials

$$\exp(tN) = \exp\left(\sum_a tN_a\right) = \prod_a \exp(tN_a) = \bigotimes_a \exp(tN_a).$$

□

Corollary (2.3). *If $\sum_i (\dim(V_i) - 1) < p$, the action $\exp(tN)$ of $\mathbb{G}_a$ on $\bigotimes_i V_i$ is the tensor product of the actions of $\mathbb{G}_a$ on the V_i defined by the N_i .*

Proof. Take $v_i = \dim(V_i)$ in 2.2. □

By the same arguments as in [4] §4, one deduces from [4] §4 prop. 10 and from 2.3 the following proposition

2.4. Let $(V_i)_{i \in I}$ be a finite family of finite-dimensional vector spaces over k , and let G be a subgroup scheme of $\prod_i \text{GL}(V_i)$. The subgroup scheme $\prod_i \text{GL}(V_i)$ of $\text{GL}(\bigoplus_i V_i)$ is saturated and infinitesimally saturated. The double saturation G^* of G , viewed as a subgroup of $\text{GL}(\bigoplus_i V_i)$, is therefore contained in $\prod_i \text{GL}(V_i)$.

Proposition (2.4). *If $\sum_i (\dim(V_i) - 1) < p$, a subspace W of $\bigotimes_i V_i$ is stable under the action of G if and only if it is stable under the action of G^* . The representation $\bigotimes_i V_i$ of G is therefore semisimple if and only if it is semisimple as a representation of G^* .*

3. The case where k is algebraically closed

Let G be an affine group scheme over the algebraically closed field k of characteristic p , and let $(V_i)_{i \in I}$ be a finite family of representations of G .

Proposition (3.1). *If the representations V_i are semisimple, and if $\sum_i (\dim V_i - 1) < p$, then the representation $\bigotimes_i V_i$ of G is again semisimple.*

Proof. We leave to the reader the verification that this statement is implied by its special case in which one makes the additional assumptions that each V_i is simple of dimension ≥ 2 , and that $|I| \geq 2$. These hypotheses imply $p \neq 2$, and each V_i has dimension $< p$.

Replacing G by its image in $\prod_i \mathrm{GL}(V_i)$, we may assume that G is a subgroup scheme of $\prod_i \mathrm{GL}(V_i)$. By 2.4, we may further assume that, as a subgroup of $\mathrm{GL}(\bigoplus_i V_i)$, G is doubly saturated. In the rest of the proof all these properties will be assumed. Saturation of G is equivalent to saturation of G_{red} , and, by [4] §4 prop. 11, implies that the finite group $G/G^0 = G_{\mathrm{red}}/G_{\mathrm{red}}^0$ has order prime to p .

By (0.7) (ii) (iii), the group G_{red}^0 is reductive and G^0 contains a connected central subgroup M of multiplicative type such that

$$M \times G_{\mathrm{red}}^0 \longrightarrow G^0$$

is an epimorphism.

By Lemma 1.3 (ii), the V_i are semisimple as representations of G^0 . If W is an irreducible representation of G^0 , then M , being of multiplicative type and central, acts on W by a character. The representation W is therefore still irreducible as a representation of G_{red}^0 . Thus the V_i are semisimple as representations of G_{red}^0 , and, by [4], the tensor product of these representations is still semisimple as a representation of G_{red}^0 . The following lemma shows that it is semisimple as a representation of G . $\square$

Lemma (3.2). *Assume that a group scheme G is an extension of a linearly reductive group scheme A by a group scheme B . Then every representation V of G which is semisimple as a representation of B is semisimple.*

That A is linearly reductive means that all its linear representations are semisimple. A group of multiplicative type, and a finite group of order prime to p , are linearly reductive.

Proof. Let S be the sum of the irreducible subrepresentations of V . We must show that $S = V$, and for this it is enough to show that S has a complement T in V stable under G . Consider the exact sequence

$$1 \longrightarrow \mathrm{Hom}_B(V/S, S) \longrightarrow \mathrm{Hom}_B(V, S) \longrightarrow \mathrm{Hom}_B(S, S) \longrightarrow 1.$$

This is an exact sequence of representations of A . Since A is linearly reductive, the identity 1_S of S , fixed by A , lifts to a homomorphism $f : V \rightarrow S$ of representations of B which is fixed by A , i.e. to a morphism of representations of G . Its kernel is the required complement. $\square$

Corollary (3.3). *Statement 0.1 is true when k is algebraically closed.*

Proof. Let $\mathcal{T}'$ be the Tannakian subcategory of $\mathcal{T}$ generated from the V_i by the operations of tensor product, dual, and passage to a subquotient. By [3] 3.3.1.1, $\mathcal{T}'$ admits a fibre functor over k , and hence is equivalent to the category of representations of an affine group scheme over k . Another reference is [1] 6.20. $\square$

Remark 3.4. The reduction from $\mathcal{T}$ to $\mathcal{T}'$ is in fact unnecessary. I have checked (unpublished) that every Tannakian category over an algebraically closed field k is equivalent to the Tannakian category of representations of an affine group scheme over k .

4. Extension of scalars in Tannakian categories and proof of 0.1

4.1. Let k be a (commutative) field and let $\mathcal{T}$ be an abelian k -linear category. The tensor product of an object X of $\mathcal{T}$ and a finite-dimensional k -vector space V is defined as the object of $\mathcal{T}$ representing the functor

$$Y \longmapsto V \otimes \mathrm{Hom}(Y, X).$$

The choice of a basis $(e_i)_{i \in I}$ of V identifies $V \otimes X$ with the sum of a family of copies of X indexed by I .

Let k' be a finite extension of k . We define as follows the abelian k' -linear category $\mathcal{T}_{k'}$ obtained from $\mathcal{T}$ by extension of scalars to k' , the scalar-extension functor $e_{k'/k} : \mathcal{T} \rightarrow \mathcal{T}_{k'}$, and the scalar-restriction functor $r_{k'/k} : \mathcal{T}_{k'} \rightarrow \mathcal{T}$. The category $\mathcal{T}_{k'}$ is the category of objects X of $\mathcal{T}$ endowed with a k' -module structure $k' \otimes X \rightarrow X$. For X in $\mathcal{T}$,

$$e_{k'/k}(X) := k' \otimes X,$$

with its natural k' -module structure. The functor $r_{k'/k}$ forgets the k' -module structure. The functors $e_{k'/k}$ and $r_{k'/k}$ are exact, and $r_{k'/k}$ is right adjoint to $e_{k'/k}$. Example: if $\mathcal{T}$ is the category of left modules over a k -algebra

A , then $\mathcal{T}_{k'}$ identifies with the category of left modules over the k' -algebra $A' := k' \otimes_k A$, and $e_{k'/k}$ and $r_{k'/k}$ are the usual scalar-extension and scalar-restriction functors.

Assume that the objects of $\mathcal{T}$ have finite length and that the Hom-groups are finite-dimensional over k . If X is a simple object of $\mathcal{T}$, the division algebra $\text{End}(X)$, being finite-dimensional over k , is central simple over its centre, which is a finite extension of k .

Lemma (4.2). *Under these hypotheses:*

- (i) *Let X be a simple object of $\mathcal{T}$, let $D := \text{End}(X)$, and let F be the centre of D . The object $e_{k'/k}(X)$ of $\mathcal{T}_{k'}$ is semisimple if and only if $F \otimes_k k'$ is a product of fields;*
- (ii) *if Y is a semisimple object of $\mathcal{T}_{k'}$, then $r_{k'/k}(Y)$ is semisimple;*
- (iii) *if X is in $\mathcal{T}$ and $e_{k'/k}(X)$ is semisimple, then X is semisimple.*

Proof. (i) The functor $Z \mapsto \text{Hom}(X, Z)$ is an equivalence from the category of objects of $\mathcal{T}$ which are sums of copies of X to the category of finite-dimensional right D -vector spaces. The category of sums of copies of X endowed with a k' -module structure identifies with the category of finite-type right $D \otimes_k k'$ -modules. The object $e_{k'/k}(X)$ of $\mathcal{T}_{k'}$ thus corresponds to the $D \otimes_k k'$ -module $D \otimes_k k'$. It is known to be semisimple if and only if $F \otimes_k k'$ is a product of fields.

(ii) One reduces to the case where Y is simple. The sum S of the simple subobjects of $r_{k'/k}(Y)$ is non-zero and is a k' -submodule. By simplicity of Y , it coincides with $r_{k'/k}(Y)$.

(iii) X is a direct factor of $r_{k'/k}e_{k'/k}(X)$, and one applies (ii). □

4.3. Now assume that $\mathcal{T}$ is a Tannakian category over k , and endow $\mathcal{T}_{k'}$ with the tensor product over k' :

$$X \otimes_{k'} Y = \text{coker}(X \otimes k' \otimes Y \rightrightarrows X \otimes Y).$$

The scalar-extension functor $e_{k'/k}$ is compatible with tensor products and with their associativity and commutativity data. It carries the unit object to the unit object, and duals (in the sense of [1] 2.2) to duals. The category $\mathcal{T}_{k'}$ has internal Homs: if X and Y in $\mathcal{T}$ are endowed with k' -module structures, their internal Hom in $\mathcal{T}_{k'}$ is the intersection of the kernels

$$\text{Hom}(X, Y) \longrightarrow \text{Hom}(X, Y), \quad f \longmapsto \lambda f - f \lambda$$

as λ runs through a basis of k' over k . More intrinsically, it is the kernel of a morphism

$$\text{Hom}(X, Y) \longrightarrow k'^{\vee} \otimes \text{Hom}(X, Y),$$

where $k'^{\vee}$ is the k -dual of the k -vector space k' . It will be denoted $\text{Hom}_{k'}(X, Y)$.

Theorem (4.4). *If $\mathcal{T}$ is a Tannakian category over k and k' is a finite extension of k , then the category $\mathcal{T}_{k'}$, endowed with the tensor product over k' , is Tannakian over k' .*

This theorem can be deduced from Theorem 1.12 of [1], which characterizes Tannakian categories as categories of representations of affine groupoids G acting transitively on a k -scheme S ($G \rightarrow S \times_k S$ affine and faithfully flat): if $\mathcal{T}$ is such a category of representations, then $\mathcal{T}_{k'}$ identifies with the category of representations of the groupoid obtained from it by extension of scalars to k' .

Conversely, following a strategy indicated to me by Grothendieck about thirty years ago, the more elementary proof of 4.4 given below should make it possible to prove [1] 1.12 more simply.

If $k' \supset k'' \supset k$, then $\mathcal{T}_{k'}$ identifies with $(\mathcal{T}_{k''})_{k'}$. This observation reduces the proof of 4.4 to the following two special cases: k'/k separable, and k'/k inseparable of degree p . The delicate point is to prove the existence of duals, in the sense of [1] 2.2, in $\mathcal{T}_{k'}$. Once this is known, if ω is a fibre functor over the k -algebra K , then ω supplies a fibre functor ω' from $\mathcal{T}_{k'}$ to the k' -algebra $K' := k' \otimes_k K$: to X in $\mathcal{T}$, endowed with a k' -module structure, attach $\omega(X)$, with its k' -module structure. This structure makes $\omega(X)$ into a $k' \otimes K$ -module.

Proof when k' is a finite separable extension of k . If X lies in $\mathcal{T}$ and has $X^{\vee}$ as dual (in the sense of [1] 2.2), the object $e_{k'/k}(X)$ of $\mathcal{T}_{k'}$ has $e_{k'/k}(X^{\vee})$ as dual. We conclude by observing that every object Y of $\mathcal{T}_{k'}$ is a direct factor of an object of this form. More precisely, the adjunction morphism

$$e_{k'/k}r_{k'/k}(Y) \longrightarrow Y$$

is split: if Y is regarded as an object of $\mathcal{T}$, then $k' \otimes Y$ has two natural k' -module structures, one coming from k' (that of $e_{k'/k} r_{k'/k}(Y)$), the other coming from Y . The adjunction morphism is the natural morphism

$$k' \otimes Y \longrightarrow (k' \otimes Y) \otimes_{k' \otimes_k k'} k'.$$

It is split because the morphism $k' \otimes k' \rightarrow k'$ is the projection onto a direct factor.

Proof when $k' = k(a^{1/p})$. Let ω be a fibre functor over an extension K of k . The key point is the following.

Lemma (4.5). *If an object X of $\mathcal{T}$ is endowed with a k' -module structure, then this structure makes $\omega(X)$ into a $k' \otimes_k K$ -module. This module is free.*

Proof. If $K' := k' \otimes_k K$ is a field, this is clear. Otherwise a has a p -th root α in K , and the K -algebra K' is isomorphic to $K[\varepsilon]/(\varepsilon^p)$: take $\varepsilon = \sqrt[p]{a} - \alpha$. Let d be the dimension over K of $\omega(X)/\varepsilon\omega(X)$. The K' -module $\omega(X)$ is isomorphic to a sum of d monogenic modules $K[\varepsilon]/(\varepsilon^j)$, $1 \leq j \leq p$. One verifies that it is free if and only if the exterior power over K'

$$\bigwedge_{K'}^d \omega(X)$$

is free (of rank one). By exactness of ω , this exterior power over K' is the image under ω of the exterior power over k' of X , defined as the image of the antisymmetrization

$$a : \bigotimes_{k'}^d X \longrightarrow \bigotimes_{k'}^d X.$$

Replacing X by this exterior power, we are reduced to the case in which $\omega(X)$ is a non-zero monogenic K' -module.

The k' -module structure of X gives a morphism

$$k' \otimes \mathbf{1} \longrightarrow \mathrm{Hom}(X, X). \quad (4.5.1)$$

The unit object $\mathbf{1}$ is known to be simple ([2] 1.17). Hence the kernel of (4.5.1) is of the form $A \otimes \mathbf{1}$, for a vector subspace A of k' . This subspace is a non-zero ideal of k' , hence $A = 0$, and (4.5.1) is a monomorphism. Applying ω , we deduce that $\omega(X)$ is a faithful K' -module, and conclude by observing that a monogenic faithful K' -module is free. $\square$

4.6. End of the proof when $k' = k(a^{1/p})$. Let $\mathbf{1}'$ denote the unit object $e_{k'/k}(\mathbf{1})$ of $\mathcal{T}_{k'}$. For X in $\mathcal{T}_{k'}$, put

$$X^\vee := \mathrm{Hom}_{k'}(X, \mathbf{1}').$$

We show that $X^\vee$ is a dual of X in the sense of [1] 2.2. In $\mathcal{T}_{k'}$ one has an evaluation morphism

$$\mathrm{ev} : X^\vee \otimes_{k'} X \longrightarrow \mathbf{1}' \quad (4.6.1)$$

and, for every Y in $\mathcal{T}_{k'}$, a morphism

$$Y \otimes_{k'} X^\vee \longrightarrow \mathrm{Hom}_{k'}(X, Y). \quad (4.6.2)$$

The functor ω' , with values in K' -modules, transforms this morphism into

$$\omega(Y) \otimes \omega(X)^\vee \longrightarrow \mathrm{Hom}_{K'}(\omega(X), \omega(Y)). \quad (4.6.3)$$

Lemma 4.5 ensures that (4.6.3) is an isomorphism. Since ω' is exact and has the property that $\omega(Z) = 0$ implies $Z = 0$, because ω has these properties, it is conservative and (4.6.2) is an isomorphism. Put $Y = X$. Composing the evident morphism

$$\mathbf{1}' \longrightarrow \mathrm{Hom}_{k'}(X, X)$$

with the inverse of (4.6.2), we obtain

$$\delta : \mathbf{1}' \longrightarrow X \otimes_{k'} X^\vee.$$

The morphisms ev and δ satisfy the identities [1] (2.1.2) making $X^\vee$ a dual of X , because those identities become true after applying ω' .

Corollary (4.7). *Let X be an object of a Tannakian category $\mathcal{T}$ over k . If X admits a module structure over an extension k' of k , then $[k' : k]$ divides $\dim X$.*

Proof. Let ω be a fibre functor of $\mathcal{T}$ over an extension K of k , put $K' := k' \otimes K$, and regard $\omega(X)$, with the k' -module structure deduced from that of X , as a K' -module. If X is regarded as an object of the Tannakian category $\mathcal{T}_{k'}$, this K' -module is the image of X under the fibre functor ω' of $\mathcal{T}_{k'}$ deduced from ω . As the image of the object X of the Tannakian category $\mathcal{T}_{k'}$ under the fibre functor ω' , it is a locally free K' -module of constant rank, equal to the dimension of X viewed as an object of $\mathcal{T}_{k'}$. Thus

$$\dim(X) := \dim_K \omega(X) = [K' : K] \cdot \text{rang}_{K'} \omega(X) = [k' : k] \cdot \dim(X \text{ viewed as an object of } \mathcal{T}_{k'}).$$

□

4.8. Warning. If D is a division algebra over k , and non-zero X admits a D -module structure, one does not necessarily have $\dim X \geq [D : k]$. For example, take $\mathcal{T}$ to be the category of $\mathbb{Z}/2$ -graded complex vector spaces V endowed with a homogeneous antilinear automorphism j , with square 1 on V^0 and -1 on V^1 . Tensor product is over $\mathbb{C}$. This category is Tannakian over $\mathbb{R}$, and over $\mathbb{C}$ it has the fibre functor “underlying complex vector space”. An odd object identifies with a vector space over the quaternion division algebra $\mathbb{H}$. A simple odd object S has dimension 2, and $\text{End}(S)$ is isomorphic to $\mathbb{H}$.

Corollary (4.9). *If X is a simple object of a Tannakian category $\mathcal{T}$ over a field k of characteristic p , and if $\dim X < p$, then the centre of $\text{End}(X)$ is a separable extension of k .*

Proof. By 4.7, it is an extension of degree $< p$. □

4.10. Let k_1 be an algebraic extension of k . If $\mathcal{T}$ is an abelian k -linear category, the categories $\mathcal{T}_{k'}$, for k' a finite extension of k inside k_1 , form a 2-inductive system of abelian categories. I write “2-inductive” rather than “inductive”, because the scalar-extension functors

$$e_{k''/k'} : \mathcal{T}_{k'} \longrightarrow (\mathcal{T}_{k'})_{k''} = \mathcal{T}_{k''}$$

for $k' \subset k''$ satisfy $e_{k'''/k''} e_{k''/k'} = e_{k'''/k'}$ only up to canonical isomorphism; these isomorphisms satisfy a cocycle condition for

$$k' \subset k'' \subset k''' \subset k''''.$$

Define $\mathcal{T}_{k_1}$ to be the inductive 2-limit of the $\mathcal{T}_{k'}$, for k' a finite extension of k inside k_1 . Since the functors $e_{k''/k'}$ are exact, the category $\mathcal{T}_{k_1}$ is still abelian.

If $\mathcal{T}$ is Tannakian, with fibre functor ω over K , the categories $\mathcal{T}_{k'}$, for $k' \subset k_1$, are Tannakian, and ω supplies a fibre functor of $\mathcal{T}_{k'}$ over $k' \otimes K$. Passing to the inductive limit, one sees that $\mathcal{T}_{k_1}$ is Tannakian, that ω supplies a fibre functor ω_1 of $\mathcal{T}_{k_1}$ over $k_1 \otimes K$, and that the $e_{k'/k}$ give a scalar-extension functor

$$e_{k_1/k} : \mathcal{T} \longrightarrow \mathcal{T}_{k_1}$$

compatible with the tensor product.

Corollary (4.11). *Let X be an object of $\mathcal{T}$.*

(i) *If the object $e_{k_1/k}(X)$ of $\mathcal{T}_{k_1}$ is semisimple, then X is semisimple.*

(ii) *If $\dim(X) < p$ and X is semisimple, then $e_{k_1/k}(X)$ is semisimple.*

Proof. For k' a finite extension of k in k_1 , the length of $e_{k'/k}(X)$ increases with k' and is bounded by $\dim(X)$. Hence there is a finite extension k' such that the length of $e_{k''/k}(X)$ is constant for all $k'' \supset k'$.

We prove (i) for X . By 4.2 (iii), it is enough to show that, for a suitable finite extension k'' of k' inside k_1 , $e_{k''/k'}(X)$ is semisimple. Let

$$0 = X_0 \subset X_1 \subset X_2 \subset \cdots \subset X_n = e_{k'/k}(X)$$

be a filtration of $e_{k'/k}(X)$ with simple successive quotients S_i . The extension X_i of S_i by X_{i-1} becomes split over k_1 . Hence there exists a finite extension k'' of k' inside k_1 over which these extensions split. The S_i remain simple because

$$\lg e_{k''/k}(X) = \lg e_{k'/k}(X).$$

It follows that $e_{k''/k'}(X)$ is semisimple. □

4.12. Proof of 0.1. Let $\mathcal{T}$ be a Tannakian category over a field k of characteristic p , and take k_1 to be an algebraic closure $\bar{k}$ of k . Corollary 4.11 reduces 0.1 for $\mathcal{T}$ to 0.1 for $\mathcal{T}_{\bar{k}}$, proved in 3.3.

5. Tensor products of exterior powers

The following results were suggested by the referee. Theorem 0.7 makes it possible to generalize Serre's theorem [5] on semisimplicity of tensor products of exterior powers of representations to the case of objects of a Tannakian category.

Theorem (5.1). *Let $\mathcal{T}$ be a Tannakian category over a field k of characteristic $p > 0$, let $(V_i)_{i \in I}$ be a finite family of semisimple objects of $\mathcal{T}$, and let $(m_i)_{i \in I}$ be a family of integers ≥ 0 . If*

$$\sum_i m_i (\dim V_i - m_i) < p, \quad (5.1.1)$$

then the tensor product of the $\bigwedge^{m_i} V_i$ is semisimple.

Recall that the m -th exterior power of an object V of $\mathcal{T}$ is the image of the antisymmetrization

$$a : \bigotimes^m V \longrightarrow \bigwedge^m V.$$

For every fibre functor ω , one has $\omega(\bigwedge^m V) = \bigwedge^m \omega(V)$, and the dual of $\bigwedge^m V$ is $\bigwedge^m (V^\vee)$.

We shall reduce the proof of 5.1 to the special case in which, for each i , $\dim(V_i) < p$. The arguments of 3.3, 4.11 and 4.12 then reduce further to assuming that k is algebraically closed, that $\mathcal{T}$ is the category of representations of an affine group scheme G over k , and that the family of representations (V_i) is faithful, i.e. that G is a subgroup scheme of $\prod_i \mathrm{GL}(V_i)$, and hence of $\mathrm{GL}(V)$, for $V := \bigoplus_i V_i$.

After some preliminaries (5.3 to 5.10), we shall reduce in 5.11 to the case where G is doubly saturated in $\mathrm{GL}(V)$, and treat that case in 5.12.

5.2. Reduction to the case $\dim(V_i) < p$. We reduce successively to assuming that

- (1) $m_i \leq \dim V_i$: otherwise $\bigwedge^{m_i} V_i = 0$;
- (2) $0 < m_i < \dim V_i$: otherwise $\bigwedge^{m_i} V_i$ is one-dimensional and the factor $\bigwedge^{m_i} V_i$ can be omitted; this changes neither the hypothesis (5.1.1) nor the conclusion, because if L is one-dimensional, semisimplicity of X is equivalent to that of $X \otimes L$, since X and $X \otimes L$ have isomorphic lattices of subobjects;
- (3) $m_i \leq (\dim V_i)/2$: otherwise replace $\bigwedge^{m_i} V_i$ by $\bigwedge^{\dim V_i - m_i} V_i^\vee$, which differs from it only by twisting by the rank-one object $\bigwedge^{\dim V_i} V_i$;
- (4) $\sum_i m_i \geq 2$: otherwise the tensor product is 1 (if $\sum_i m_i = 0$) or is reduced to a single V_i (if $\sum_i m_i = 1$).

If these conditions hold, then $\dim V_i < p$. Indeed, if $|I| \geq 2$, (5.1.1) implies

$$\dim V_i - 1 = 1(\dim V_i - 1) \leq m_i(\dim V_i - m_i) < p - 1,$$

and if $|I| = 1$, then $m_i \geq 2$, $\dim V_i \geq 4$, and

$$\dim V_i \leq 2(\dim V_i - 2) \leq m_i(\dim V_i - m_i) < p.$$

5.3. Let V be a representation of $\mathrm{SL}(2)$. The action of the multiplicative group of diagonal matrices (a, a^{-1}) defines a grading $V = \bigoplus V^j$ of V . The action of $\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix} \in \mathrm{SL}(2)$ exchanges V^j and V^{-j} . Hence $\dim V^j = \dim V^{-j}$. Let E denote the action of the element $\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$ of the Lie algebra. This endomorphism of V has degree 2. If the weights of V are $< v$, i.e. if $V^j = 0$ for $|j| \geq v$, then $E^v = 0$.

If the weights of V are $< p$, it is known that the representation V is semisimple, a sum of Sym^i of the fundamental representation with $i < p$, and that the j -th iterate E^j of E induces an isomorphism

$$E^j : V^{-j} \xrightarrow{\sim} V^j. \quad (5.3.1)$$

Conversely, if V is a graded vector space, with $V^j = 0$ for $|j| \geq p$, and if E is an endomorphism of degree 2 of V satisfying (5.3.1), then there is a unique action of $\mathrm{SL}(2)$ on V defining this grading and this endomorphism, one

has $E^p = 0$, and the action of $\begin{pmatrix} 1 & t \\ 0 & 1 \end{pmatrix}$ in $\mathrm{SL}(2)$ is given by the truncated exponential

$$\exp(tE) := \sum_{n < p} \frac{(tE)^n}{n!}. \quad (5.3.2)$$

5.4. Let E be a nilpotent endomorphism of a vector space V of dimension $\leq p$. Decomposing E into Jordan blocks, one checks that E comes from a representation of $\mathrm{SL}(2)$ as in 5.3, with weights $< p$. If E has only one Jordan block, the dominant weight is $\dim(V) - 1$.

Lemma (5.5). *With the hypotheses and notation of 5.4, the weights of the representation $\bigwedge^m V$ of $\mathrm{SL}(2)$ are $\leq m(\dim V - m)$.*

Lifting the representation V of $\mathrm{SL}(2)$ to characteristic 0, one is reduced to verifying in characteristic 0 that, for a representation W of $\mathrm{SL}(2)$ of dimension d , one has

$$\text{the weights of the representation } \rho_m \text{ of } \mathrm{SL}(2) \text{ on } \bigwedge^m W \text{ are } \leq m(d - m). \quad (5.5.1)$$

Since, in characteristic 0, every representation of $\mathrm{SL}(2)$ is a direct sum of Sym^j of the fundamental representation, (5.5.1) is equivalent to

$$(d\rho_m(\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}))^{m(d-m)+1} = 0. \quad (5.5.2)$$

Let E be the nilpotent endomorphism of W giving the action of $\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \in \mathfrak{sl}(2)$. If one identifies endomorphisms and actions of the one-dimensional Lie algebra, then $d\rho_m(\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix})$ is the natural extension E_m of E to $\bigwedge^m W$. Nilpotent endomorphisms with a single Jordan block are dense among nilpotent endomorphisms. It is therefore enough to verify that $(E_m)^{m(d-m)+1} = 0$ when E has only one Jordan block. In that case, the weights of the representation W are $(d-1), (d-3), \dots, -(d-1)$, each with multiplicity one. The largest weight of the representation $\bigwedge^m V$ is thus

$$(d-1) + (d-3) + \dots + (d-(2m-1)) = md - m^2 = m(d-m),$$

and (5.5.2) follows.

5.6. Return to characteristic p . Let N be a nilpotent endomorphism of a vector space V of dimension $d \leq p$, and let N_m be its extension to $\bigwedge^m V$ (in the Lie-algebra sense, as in 5.5). Since $d \leq p$, one has $N^p = 0$, and N defines an action $\rho(t) = \exp(tN)$ of $\mathbb{G}_a$ on V . This action extends to an action of $\mathbb{G}_a$ on $\bigwedge^m V$.

Proposition (5.7). *With the notation of 5.6, one has*

$$(N_m)^{m(d-m)+1} = 0 \quad (5.7.1)$$

and if $m(d-m) < p$, the action of $\mathbb{G}_a$ on $\bigwedge^m V$ is given by the truncated exponential $\exp(tN_m)$.

Proof. As in 5.4, there exists an action ρ , with weights $< p$, of $\mathrm{SL}(2)$ on V such that $d\rho(\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}) = N$. It extends to an action ρ_m of $\mathrm{SL}(2)$ on $\bigwedge^m V$. By 5.5, $\bigwedge^m V$ has weights $\leq m(d-m)$. The vanishing (5.7.1) follows. If $m(d-m) < p$, by 5.4 the action of $\begin{pmatrix} 1 & t \\ 0 & 1 \end{pmatrix} \in \mathrm{SL}(2)$ on $\bigwedge^m V$ is given by the truncated exponential $\exp(tN_m)$. $\square$

5.8. Let (V_i) be a finite family of vector spaces of dimension $< p$, and let (m_i) be a family of integers ≥ 0 . For each i , let N_i be a nilpotent endomorphism of V_i . The truncated exponential $\exp(tN_i)$ is an action of $\mathbb{G}_a$ on V_i . These actions define an action of $\mathbb{G}_a$ on $\bigotimes_i \bigwedge^{m_i} V_i$, and, by passage to the Lie algebra, an endomorphism N of this tensor product. It is deduced from the N_i , viewed as actions on the V_i of the one-dimensional Lie algebra.

Corollary (5.9). *With the notation of 5.8, if*

$$\sum_i m_i(\dim V_i - m_i) < p, \quad (5.9.1)$$

then $N^p = 0$, and the action of $\mathbb{G}_a$ on $\bigotimes_i \bigwedge^{m_i} V_i$ is given by the truncated exponential $\exp(tN)$.

Proof. As in 5.4, choose an action ρ_i of $\mathrm{SL}(2)$ on V_i , with weights $< p$, such that $d\rho_i\left(\begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}\right) = N_i$. By 5.5, the weights of $\mathrm{SL}(2)$ acting on $\bigotimes_i \bigwedge^{m_i} V_i$ are $\leq \sum_i m_i(\dim V_i - 1) < p$, and one applies 5.4 as in 5.7. One could equally well apply 5.7 and 2.2. $\square$

5.10. The truncated exponential induces a bijection between nilpotent endomorphisms N such that $N^p = 0$ and automorphisms g such that $g^p = 1$, and the corresponding actions of $\mathbb{G}_a$ coincide: $\exp(N)^t = \exp(tN)$. With the notation of 5.8, if the automorphism g_i of V_i satisfies $g_i^p = 1$, and g is the automorphism of $\bigotimes_i \bigwedge^{m_i} V_i$ induced by the g_i , condition (5.9.1) ensures that g^t is induced by the g_i^t .

Recall that, by the remarks following 5.1, 5.2 reduces the proof of 5.1 to the case in which k is algebraically closed and $\mathcal{T}$ is the category of representations of $G \subset \prod_i \mathrm{GL}(V_i) \subset \mathrm{GL}(V)$, with $V := \bigotimes_i V_i$.

5.11. Reduction to the case where G is doubly saturated in $\mathrm{GL}(V)$. Let G^* be the double saturation of G (2.1). As in 2.4, 5.9 and 5.10 ensure that, if (5.1.1) holds, a subspace of $\bigotimes_i \bigwedge^{m_i} V_i$ stable under G is automatically stable under G^* . Semisimplicity as a representation of G^* therefore implies semisimplicity as a representation of G .

5.12. End of the proof of 5.1. As in 3.1, one reduces to assuming G reduced. One can then apply [5] p. 26, a consequence of the “Main Theorem” p. 20, to the representations V_i of $G(k)$. One could instead invoke [6].

5.13. In 5.1, we use the Schur functors $V_i \mapsto \bigwedge^{m_i} V_i$, associated with the representations of the $\mathrm{GL}(\dim V_i)$ of dominant weight $(1, \dots, 1, 0, \dots, 0)$ (m_i “1”s). More generally, one can consider Schur functors $S_{\lambda(i)}$, where $\lambda(i)$ is a dominant weight $(\lambda(i)_1, \dots, \lambda(i)_{\dim V_i})$ of $\mathrm{GL}(\dim V_i)$. Regard $\lambda(i)$ as a partition. Let $\mu(i)$ be its dual and put

$$n(\lambda(i)) = \sum_j \mu(i)_j (\dim V_i - \mu(i)_j)$$

(cf. [6] 5.2). Condition (5.5.1) is to be replaced by

$$\sum_i n(\lambda(i)) < p. \quad (5.13.1)$$

That (5.13.1), together with semisimplicity of the V_i , suffices to ensure semisimplicity of $\bigotimes_i S_{\lambda(i)} V_i$ can be deduced from 5.1 and from the fact that $S_{\lambda(i)}$ is a subquotient (and even, under the hypotheses made, a direct factor) of the functor

$$V \mapsto \bigotimes_j \bigwedge^{\lambda(i)_j} V$$

(for V of dimension $\dim(V_i)$).

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